Raj Patel

PhD Student · Department of Computer Science · The University of Alabama

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RESEARCH INTERESTS

AI security and secure MLOps; agentic AI for enterprise intrusion detection and response; federated and privacy-aware intrusion detection; healthcare, wearable and medical-device security.

EDUCATION

Ph.D., Computer Science, The University of Alabama Fall 2025 – present

Advisor: Dr. Shahram Rahimi · GPA 4.000/4.000 · Transferred 18 graduate credits from Mississippi State through individual course equivalencies

Doctoral study, Computer Science, Mississippi State University Spring 2024 – 2025

Transferred to The University of Alabama in Fall 2025

M.S., Computer Science, Mississippi State University August 2022

GPA 4.00/4.00

B.S., Computer Science, Mississippi State University December 2018

GPA 3.79/4.00

A.S., Hinds Community College December 2015

GPA 3.81/4.00

PUBLICATIONS

Journal Articles

[J1] **R. Patel**, H. Tripathi, J. Stone, N. Amiri Golilarz, S. Mittal, S. Rahimi, and V. Chaudhary. "Towards Secure MLOps: Surveying Attacks, Mitigation Strategies, and Research Challenges." *ACM Transactions on AI Security and Privacy*, 2026. Accepted December 30, 2025. doi:10.1145/3796728

Recognized by the Editors-in-Chief as the first manuscript accepted at the journal.

Refereed Conference and Workshop Papers

[C8] M. Guida, S. Iannucci, **R. Patel**, S. Rahimi, S. Mittal, and P. Merialdo. "Shortcut Learning in Early Intrusion Detection: Header Fingerprints Across Four Datasets." *8th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS)*, 2026. Code: github.com/micheleguidaa/early-nids-artifact

Best Student Paper Award.

[C7] S. Mitra, **R. Patel**, S. Neupane, S. Mittal, M. R. Rahman, and S. Rahimi. "Where Rules End and Judges Begin: Measuring the Judgment Boundary in Multi-Agent Systems Security." *Workshop on Agentic AI in Offensive and Defensive Cyber Operations (AIDC '26), co-located with ACSAC 2026*. Accepted. arXiv:2610.07657

[C6] **R. Patel**, S. Mitra, M. Guida, S. Iannucci, S. Mittal, and S. Rahimi. "Agentra: A Supervisable Multi-Agent Framework for Enterprise Intrusion Response." *2026 International Conference on Machine Learning and Applications (ICMLA)*. Accepted. arXiv:2606.18325

[C5] S. Mitra, **R. Patel**, S. Mittal, M. R. Rahman, and S. Rahimi. "AgenticCyOps: Securing Multi-Agentic AI Integration in Enterprise Cyber Operations." *2026 International Conference on Machine Learning and Applications (ICMLA)*. Accepted. arXiv:2603.09134

[C4] **R. Patel**, S. Mitra, D. Amebley, T. Akinrele, S. Dibbo, S. Mittal, and S. Rahimi. "The Value of Spike Timing: A Leakage-Resistant Benchmark of SNN Design Choices for Network Intrusion Detection." *8th IEEE International Conference on Cognitive Machine Intelligence (CogMI)*, 2026. arXiv:2606.01442

Best Student Paper Award.

[C3] D. Panigrahi, **R. Patel**, S. Mitra, S. Mittal, and S. Rahimi. "IRSDA: An Agent-Orchestrated Framework for Enterprise Intrusion Response." *2025 IEEE International Conference on Big Data (BigData)*, pp. 6616–6625. doi:10.1109/BigData66926.2025.11402393

[C2] T. Pawelek, **R. Patel**, C. Crowell, N. Amiri Golilarz, S. Mittal, S. Rahimi, and A. Perkins. "LLMZ+: Contextual Prompt Whitelist Principles for Agentic LLMs." *2025 International Conference on Machine Learning and Applications (ICMLA)*, pp. 1396–1402. doi:10.1109/ICMLA66185.2025.00213

[C1] J. Stone, **R. Patel**, F. Ghiasi, S. Mittal, and S. Rahimi. "Navigating MLOps: Insights into Maturity, Lifecycle, Tools, and Careers." *2025 IEEE Conference on Artificial Intelligence (CAI)*, pp. 643–650. doi:10.1109/CAI64502.2025.00118

Under Review

[R3] Y. Dai, S. Mitra, **R. Patel**, Y. Sun, S. Chen, J. Gong, S. Mittal, S. Rahimi, X. Jia, and R. Yu. "Staying on the Attractor: Supervising Neural Surrogates of 3D Turbulence Where They Leave It." Under review. arXiv:2609.32864, 2026.

[R2] S. Mitra, **R. Patel**, S. Mittal, M. R. Rahman, and S. Rahimi. "Agentic AI for Cyber Defense: A Survey of Failure Modes, Applications, and Research Opportunities." *ACM Transactions on AI Security and Privacy*, minor revision. Preprint: Preprints.org 202608.0865, 2026.

[R1] **R. Patel**, U. Biswas, S. Kodipaka, W. Carroll, P. Peranich, and M. Young. "A Survey of Recent Advancements in Secure Peer-to-Peer Networks." *ACM Computing Surveys*, under review. arXiv:2509.19539, 2025.

RESEARCH EXPERIENCE

Graduate Researcher, Department of Computer Science, The University of Alabama

Fall 2025 – present

Advisor: Dr. Shahram Rahimi

- Submitted the Secure MLOps survey to ACM TAISAP and led the major revision through acceptance [J1].
- Led a leakage-resistant benchmark of spiking neural networks for network intrusion detection [C4]; developing a journal extension with Dr. Sayanton Dibbo.
- Led Agentra, a supervisable multi-agent intrusion response framework (ICMLA 2026) [C6], and contributed to work on securing multi-agent AI in cyber operations [C5, C7, R2].
- Developing a conference paper on adversarial machine learning with Dr. Mahmoud Mahmoud.

Doctoral research, Mississippi State University

Spring 2024 – 2025

- Began the Secure MLOps survey; first preprint posted May 2025.
- Co-authored a unified MLOps lifecycle framework covering maturity, tools and careers [C1].

Student Researcher, Center for Advanced Vehicular Systems, Mississippi State University

Mar 2017 – May 2018

- Developed C++ communication software over TCP/UDP, I²C, SPI and UART for microcontroller systems.
- Built a Python hardware-in-the-loop simulation for vehicle-system testing.

PROFESSIONAL EXPERIENCE

Systems Analyst II, Department of Computer Science, The University of Alabama

May 2025 – present

- Plan and configure software systems that support the department's administrative functions, including departmental and moderately critical applications.
- Install, configure, support and maintain integrations for vendor-supplied software systems.
- Collaborate with departments, colleges and campus customers to troubleshoot and resolve issues promptly.

Assistant Systems Administrator, Mississippi State University

May 2019 – May 2025

- Installed and configured 25+ servers and 300 workstations, reducing setup time by 25% through improved deployment processes.
- Managed 1,100+ user accounts and permissions; automated Active Directory account creation for 5,000 students with PowerShell.
- Managed patches in response to vulnerability reports and enforced security controls through Group Policy.
- Designed virtual-server, storage and network solutions; implemented a virtual desktop infrastructure with vGPU for 300 concurrent users, saving about $25K a year with open-source software; deployed an OpenStack environment for teaching.
- Developed a C# tool that cut Crestron NVX installation time by 50% and a Laravel course-override application that cut related email by 40%.
- Supervised and trained four student workers; coordinated computing infrastructure and procurement with research laboratories and vendors; audited university and federally sponsored assets.

Junior Software Developer, BeyondTrust Corporation

May 2018 – Aug 2018

- Developed remote-support shell-sharing software in C++ with a Qt interface and multithreaded execution.
- Worked with quality assurance on testing and documentation; used Git and Bitbucket in an Agile process.

TEACHING & MENTORING

Teaching preparation, The University of Alabama

Fall 2026

- AHE 603, College and University Teaching.
- Teaching portfolio and course portfolio for CS 456, AI for Cyber Security: course goal, objectives and outcomes, course map, assessments and a sample lesson (rajpatel-ua.github.io/teaching.html).

Student-worker supervisor, Mississippi State University

2019 – 2025

- Trained and supervised four student workers; improved team task efficiency by 40% through documentation and access-management procedures.

SERVICE

Department service, Mississippi State University 2019 – 2025

- Managed and rebranded the department's social media on Facebook, LinkedIn, X and Instagram, with student designers.
- Helped plan departmental social events.
- Worked with research labs, including Esports, PATENT, AIR, Hi5, RAPTOR and Cybersecurity, on their technology and facility needs.

HONORS & RECOGNITION

Best Student Paper Award, IEEE CogMI 2026 (first author) November 2026

For [C4], "The Value of Spike Timing" · San Jose, CA

Best Student Paper Award, IEEE TPS 2026 (co-author) November 2026

For [C8], "Shortcut Learning in Early Intrusion Detection" · San Jose, CA

First manuscript accepted at ACM Transactions on AI Security and Privacy December 2025

Recognized by the journal's Editors-in-Chief in the acceptance decision for [J1]

GRADUATE COURSEWORK

The University of Alabama	Adversarial Machine Learning; Computer Algorithms; Physics-Guided Machine Learning; Independent Study (two terms). In progress: College and University Teaching; Mentored Research Experience.
Transferred from MSU	Artificial Intelligence; Distributed Computing; Cognitive Skill Models; Advanced Cyber Operations; Data Science Concepts; Wireless Networks.

TECHNICAL SKILLS

AI & security	Adversarial machine learning, multi-agent LLM systems, spiking neural networks, intrusion detection and response; MITRE ATT&CK, ATLAS and D3FEND; NIST CSF 2.0.
Programming	Python, C++, C#, PowerShell, Laravel, Qt; Git, Bitbucket, Visual Studio, WordPress.
Infrastructure	Active Directory, Group Policy, patch management, server and workstation deployment, virtual desktop infrastructure (vGPU), OpenStack, virtualization, storage and network administration.
Leadership	Student-worker supervision, technical documentation, vendor coordination, procurement and asset management.

REFERENCES

Available upon request.

Last updated November 2026